

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{M}_{xy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzzxy} \\ 0 & 0 & 0 & 0 & M_{yzzzx} & 0 \\ 0 & 0 & 0 & M_{yzzzx} & 0 & 0 \\ M_{xxxxy} & M_{xxxxy} & M_{zzzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{zzzxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{yzzzx} = \mathbb{M}_{xy}^{(2)} + 2\mathbb{M}_z^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{xxxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxxy} & -M_{xxxxy} & -M_{zzzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxxy} = \mathbb{M}_{xy}^{(3)} + 2\mathbb{T}_z^{\beta(1)}$$

$$M_{zzzxy} = \mathbb{M}_{xy}^{(3)} - 4\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzzx} & 0 \\ M_{yzzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = -2\mathbb{M}_{xy}^{(5)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{yzzx} = -\mathbb{M}_{xy}^{(5)} - \mathbb{T}_z^{\beta(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{xxyx} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)} + 2\mathbb{T}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} - 4\mathbb{T}_z^{\beta(1)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} - \mathbb{T}_z^{\beta(2)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)} + \mathbb{T}_z^{\beta(2)}$$

$$M_{xyxx} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)} - 2\mathbb{T}_z^{\beta(1)}$$

$$M_{xyyz} = \mathbb{M}_{xy}^{(1)} - \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)} + 4\mathbb{T}_z^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = 2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{yzzx} = -2\mathbb{M}_{xy}^{(4)} + \mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

$$M_{xyxx} = -2\mathbb{M}_{xy}^{(6)} + \mathbb{T}_z^{\beta(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & M_{xxxy} & 0 \\ M_{xxxy} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{zzxy} & 0 \\ M_{xxyx} & 0 & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$M_{xxxy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} - 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{zzxy} = \mathbb{M}_{xy}^{(1)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{xxyx} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} + \mathbb{M}_{xy}^{(3)} + 2\mathbb{M}_{xy}^{(5)} - \mathbb{M}_z^{\beta(1)}$$

$$M_{yzzx} = \mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(5)} + 2\mathbb{M}_z^{\beta(1)}$$

$$M_{xyyy} = \mathbb{M}_{xy}^{(1)} + 2\mathbb{M}_{xy}^{(2)} - \mathbb{M}_{xy}^{(3)} - \mathbb{M}_z^{\beta(1)}$$